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UNIVERSITI TEKNOLOGI MALAYSIA

FACULTY OF COMPUTING
UTM Johor Bahru

Project : Phase 2

<Minisite.com>

SECD2523 - SYSTEM ANALYSIS AND DESIGN

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1.0 Overview of project

We will collect information requirements and look at system requirements for this project. This project will determine the information needs that can help you grasp the 4W1H of an operational system. We use Google Forms to poll customers engagingly to gather information from them. We use nominal and scale queries to elicit more detailed data from various points of view. After gathering the requirements information, we have a solid understanding of the business function based on a piece of precise information about the people, objectives, data, and procedures involved.

The business flow of the AS-IS system for the e-Commerce Business is then visualised using context diagrams in the following steps which are described as a high-level Data Flow Diagram (DFD), a context diagram depicts the information flow between the minisite.com Business system and outside entities, including sellers and couriers.

Data flow diagrams, including the parent and child diagrams, are used to depict the external entities, data flows, data stores, and procedures involved once the context diagram has been created. The explosion of context diagram, often known as the parent diagram, is a level 0 diagram. An integer is assigned to each process. The parent diagram includes all external entities as well as the main data storage. Each process in the parent diagram will be exploded to generate a child diagram. A child diagram utilises the same number as the parent diagram, receives the same input, and produces the same output.

After DFD, the Entity Relationship Diagram (ERD), which depicts the relationships for an entity set, is constructed. After the ERD, data requirements and transaction requirements are generated. While the transaction requirements give some examples of how the data is used, the data requirements show how the data is used.

The tasks are finally completed, and we now thoroughly understand the current system and data flow diagrams, entity relationship diagrams, and information requirements.

2.0 Problem Statement

1. Hard to analyse order data and product status.

The stakeholder, Mrs. Chelsea needs to know the current status to update manually the order data record, she was facing problems and felt hard because she always spent her a lot of time doing analysis based on the sales. Besides that, she also found it hard to compare the sales based on the order data and the price with other companies due to the unsorted order data not being organised. Due to the inadequate analysed order data supply on time, she always missed catching up with the business timing since she won't know which items were popular and in people's demand by predicting through the data.

2. Hard to track the parcel that shipped out.

Mrs. Chelsea was having difficulties in tracking where the parcel goes and hard to estimate the time when the parcel reaches the customer since she was responsible for informing the customers regarding the product's information when the orders had been made successfully. Sometimes customers may place orders in advance and set to receive the deliveries at a specific time send to a particular place, thus Mrs. Chelsea and customers are unable to know where the parcel is located exactly and when the parcel will be shipped out, which courier services were responsible in shipping the parcel and even the time when the parcel arrives. Besides that, while tracking the parcel, the stakeholder also will need to know the order details, which can allow the user to determine the detail of the product's order, including the product purchased, order ID, and shipping address. Since she could not know track which product numbers were sold, in transit, or on delivery, it brings her trouble indirectly in analyzing the order of the product and the stock available.

3. Hard to check and update the stock of products.

Mrs. Chelsea was required to check the stock and inventory manually every day to ensure that there is a sufficient available stock that can be supplied on time to her business. While checking the stock, she found that it was hard to count the quantity of the product that still owns currently, but the problem is that she did not have enough time to always check the product stocks and the status one by one because of her limited ways must be done manually. In addition, Mrs. Chelsea was hard to update the product descriptions for more detailed information because she needed to update them one by one manually and repeat the process every time since there were too many product descriptions that needed to mention for each type of product, for example, retail price, the available stock, and product information.

Sometimes, she needs to check and predict the stock in advance and ensure a sufficient stock to allow the customers to place orders because some customers will add particular items to their carts before making the payment. Thus, updating the available stock information on time was hard on the online platforms. It was hard to know the stock listing since it is hard to list all the products published on online platforms and the details of products, including single products and bundle products.

4. Hard to check the order according to the term.

The problem causes her to have difficulties in analysing the order since when she needs to analyse the order data based on the product, she found that she needs to search for the specific product from the unsorted data, but she was unable to do so. Not only this, but when she wants to deliver the order, she needs to view the previous history order, she was unable to select to filter out all the unrequired data. Due to the limitations, she was hard and inefficient in completing her work because it always causes her to spend a lot of time searching for the product.

5. Hard to check the online payment status.

The problem causes her to have difficulties in analysing the order since when she needs to analyse the order data based on the product, she found that she needs to search for the specific product from the unsorted data, but she was unable to do so. Not only this, but when she wants to deliver the order, she needs to view the previous history order, she was unable to select to filter out all the unrequired data. Due to the limitations, she was hard and inefficient in completing her work because it always causes her to spend a lot of time searching for the product.

3.0 Proposed Solution

Minisite is a new system solution that we are trying to propose and introduce to the stakeholder, Mrs. Chelsea, in which we are concentrating on bringing convenience to the business. It is a system that provides a way to analyse the order data which make analysis processes to be easier because it will collect the data in organized form and list them based on the various type of products in terms of product ID, product type, brand, available stock, and so forth. This system will categorize the product based on the product status, defective, quantity, product description, and so forth. This system also provides an easy and free navigation method interface.

This system will provide airway bill printing, allowing users to print the airway bill according to the order and the customers can be tracked easily based on the tracking code provided by the courier services. Mrs. Chelsea able to know which number of product types with the specific number that will be shipped out previously, pending for shipping, and the product that will be shipping in the future. For example, the type of product that had been chosen by the customer will be stated along with the product numbers and delivery address, the date and the time of the product being shipped out and reaching the destination, and customer information clearly to help her identify her products has been successfully delivered to the customers.

Previously, it was hard to know the latest status of the product since she needs to update all the records manually. Thus, it was hard to synchronize all the order data and update it immediately. This system tried to provide and show the latest information to the stakeholder using the technique and the analyzing order's data method. This system will always update the stock available on time once the quantity of the product changes and the status of the product, thus it will increase the efficiency in saving time in checking and counting the product from numerous product statuses.

Besides that, this system will also provide Mrs. Chelsea with the product listing which state all the available product stock in a neat way which aims to help her in checking the stock and updating the stock. A sufficient organized order data supply could help the stakeholder catch up on the chance and predict the trend based on updating of the order data and the stock earlier. In addition, this system provides stock management for the user, enabling the stakeholder to make a stock

adjustment. It provides a stock listing to list all the products published on online platforms and the details of the products.

Furthermore, Minisite was able to provide a way for the stakeholder Mrs. Chelsea to check the order according to their preferences. Firstly, this system required the stakeholder, Mrs. Chelsea, to log in advance and customize the data according to their interests and preferences. Hence, all the product data collected will be analysed, organized easily, and listed in the proper form based on the products. The customizing method allows the stakeholder to know where the data is located easily and how the data is being categorized. By searching the term, this system will show all the related data according to the term to allow her to select and filter out all the unrequired data to make the process easier. Not only this, but for user-friendly consideration, this system is designed to allow the user to select the date, either based on the period or the specific date to show the recent history of product data easily.

Moreover, this system provides a function of notification that can help stakeholders to be alert when there is any transformation or the payment had been made successfully by sending an email to inform Mrs. Chelsea. For example, a message regarding the defective product should be stated clearly along with the product ID to indicate the particular product exactly and a notification regarding the amount of payment that had been made by the customer successfully. Thus, she can easily identify the defective product to avoid being delivered to the customers, which lowers the business rating and avoids miss sending the products to the customer. Hence, this system can help in avoiding checking manually whether the payment was made successfully or not by the customers while using the online transfer method since this system will update immediately and notify them once the customer has made the payment successfully, along with a clear product description. Last but not least, when the product's stock was reaching the alert value, this system will try to send a notification to inform the stock status.

4.0 Information Gathering Process

Information collecting is important to learn as much as possible about the AS-IS system. These facts were acquired from Mdm. Chelsea, a shareholder in our project based on the AS-IS system, and the 100 sellers who used minisite.com as users.

4.1 Method Used

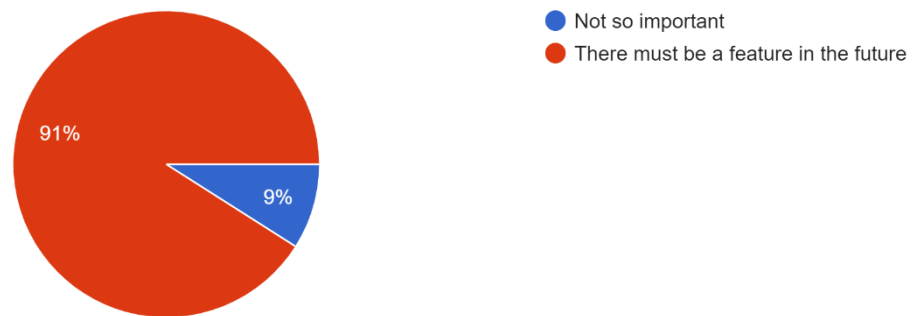
To learn more about the AS-IS system, we employed an interview and questionnaire containing both closed- and open-ended design questions. Google Forms, an online platform, is used to conduct the survey. The questions, 100 sellers' responses, and Mdm. Chelsea's comments are all presented here.

4.1.1 Questionnaires

Question 1: Do you think that the analysis of order data going easy is important?

1. Do you think that the analysis of order data going easy is important?

100 responses

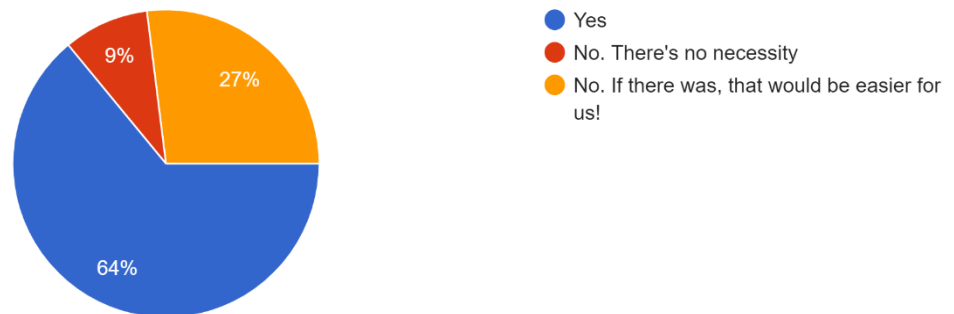


According to the responses to the questionnaire, the majority of sellers consider that the analysis of order data going easy is very important in the minisite.com system. It is because there will be trouble when the order data is unable to be read clearly.

Question 2: 2. Do you know the journey of the parcel that you shipped out? If no, do you need it?

2. Do you know the journey of the parcel that you shipped out? If no, do you need it?

100 responses

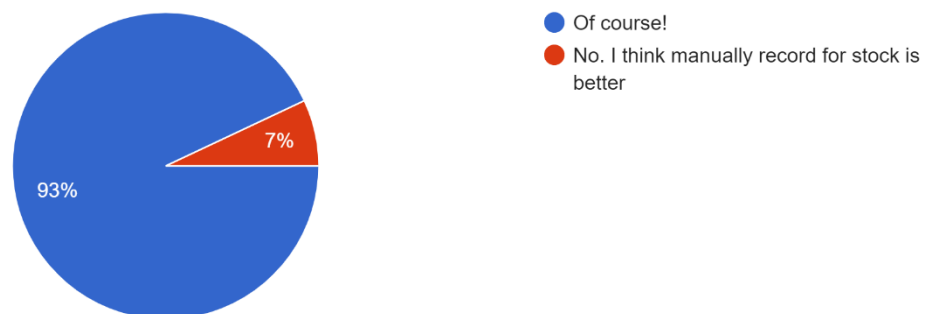


Based on the answer to the second question in the survey form. 27% of sellers don't know about the journey of the parcel that they're shipped out. It is because the parcel just will be updated by the courier partner when arriving at one point, and is unable to get the real-time update. When these features are added to minisite.com, that would be easier for them.

Question 3: Do you think that the stock system is necessary?

3. Do you think that the stock system is necessary?

100 responses

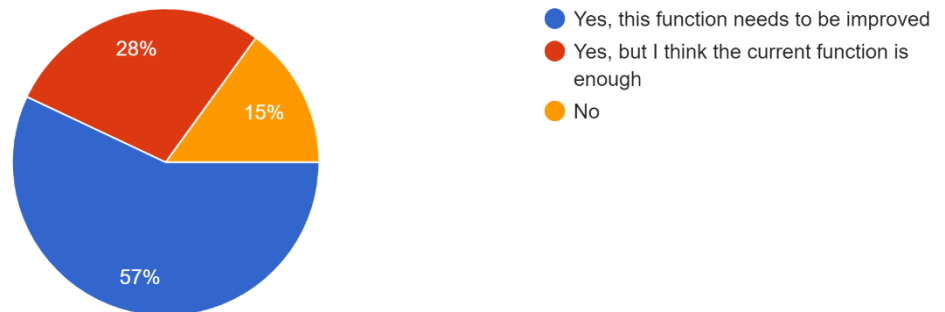


Based on the data that get from the questionnaires, many of them agree that the stock system is necessary for them. So, the stock system is a part of the online merchant system which can help them easily to update and track the inventory.

**Question 4: Do you think that the status of online payment is difficult to track?
If yes, is it necessary to improve it?**

4. Do you think that the status of online payment is difficult to track? If yes, is it necessary to improve it?

100 responses

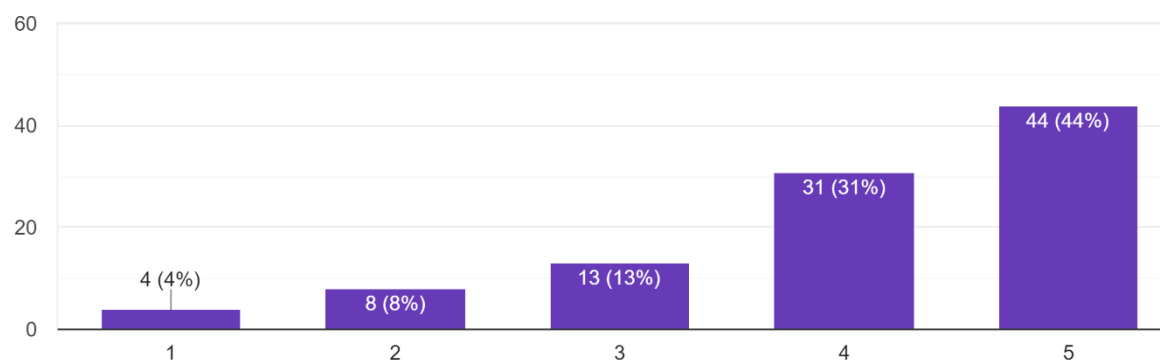


Based on the question stated in this survey form, 57% of the users are thinking that the online payment status is difficult to track, and the users need to wait for some working days, the payment status just will be updated in their system. Meanwhile, they are urged to improve this function to make their selling progress smoothly.

Question 5: View order according to preferences

5. View order according to preferences

100 responses



Based on the question, the responders were given a scale of 1 to 5 with 1 indicating not need and 5 indicating would be very helpful for them. The response that was provided on the questionnaire stated the majority of users answered on a scale of 5 out of 5, which is very needed to have special instruction on minisite.com. This is because the current system lacks that capability.

4.1.2 Interviewing

Question 1: How do you check the payment status on the current system?

1. How do you check the payment status on the current system?

1 response

I need to log in my shopee account and I need to go to sales and check for the payment

The interview shows that Mdm. Chelsea needs to log in to her online selling account to check about the payment status. This means Mdm. Chelsea is unable to track the payment status on a designed page.

Question 2: How do you track the parcel that you shipped out to your buyers?

2. How do you track the parcel that you shipped out to your buyers?

1 response

I need to log in to my shopee account and I need to press the order one by one to check the tracking status

Same as the previous question, Mdm. Chelsea needs to log on to her online selling account at a specific online shopping platform, just can check the parcel status. If Mdm. Chelsea is having more than one account on different online shopping platforms, she needs to log on to the different accounts to check the customer's parcel status, which makes longer time on the checking progress.

Question 3: In the current system, how do you analyse the sales order by the customer and what improvement about sales order can be done.

3. In the current system, how do you analyse the sales order by customer and what improvement about sales order can be done.

1 response

I need to manually key in the sales order into the system. I hope that the system will be able to auto-sync the sales orders and generate a monthly sales report for me

Based on the response in the interview form, Mdm. Chelsea is arranging the inventory record by using Google Sheets or Microsoft Excel, which might need her to key in herself to make the inventory record.

Question 4: If you want to search the order by month, how do you search the order on the current system?

4. If you want to search the order by month, how you search the order on the current system?

1 response

I need to open a new sheet for every month and when I would like to search the order per month, I could only check one by one together with the sales order in my shopee account and when I need to check a particular kind of product, I need to manually list out the order ID and analyze on my own.

The response shows that the current system cannot let Mdm. Chelsea to sort the order history by her preferences. She needs to extract the order history from a file and analyse it herself. When Mdm. Chelsea needs to track the number of orders for a particular product, she is also unable to get it easily from the current system.

Question 5: What do you think the weakest of the current system?

5. What do you think the weakest of the current system?

1 response

Non automated system is costing a lot of my time

The response shows that the current system is not automated, which needs the users to do something manually. It will cause the users to lose most of the time on the analysis of order data, tracking of payment status and package delivery path. Besides that, the users also need to waste theirs on the analysis of the order number for specific products to make sure it is gaining profit.

Question 6: How can improve the weakest of the current system?

6. How can improve the weakest of the current system?

1 response

A more automated system will be a big plus for me

From the interview form, Mdm. Chelsea hopes that the system can be innovated into a more automated system to make her selling progress smooth and functional.

4.2 Summary of Method Used

We were able to learn more about how our stakeholder business system functions from the perspectives of the stakeholder and other online selling users, which helped us better understand how we can design a system that enables Richmine Trading to operate more effectively and efficiently. We concentrate more on developing a system that enables us to more effectively and automatically gather consumer interests based on their orders from the questions we ask via a questionnaire. After receiving responses to the survey, we were able to decide which components of the current system we ought to prioritise. For example, keeping track of the delivery guy's live location as well as the estimated delivery time is essential for keeping customers and exceeding their expectations.

3.0 Requirement Analysis

3.1 Current business process (scenarios, workflow)

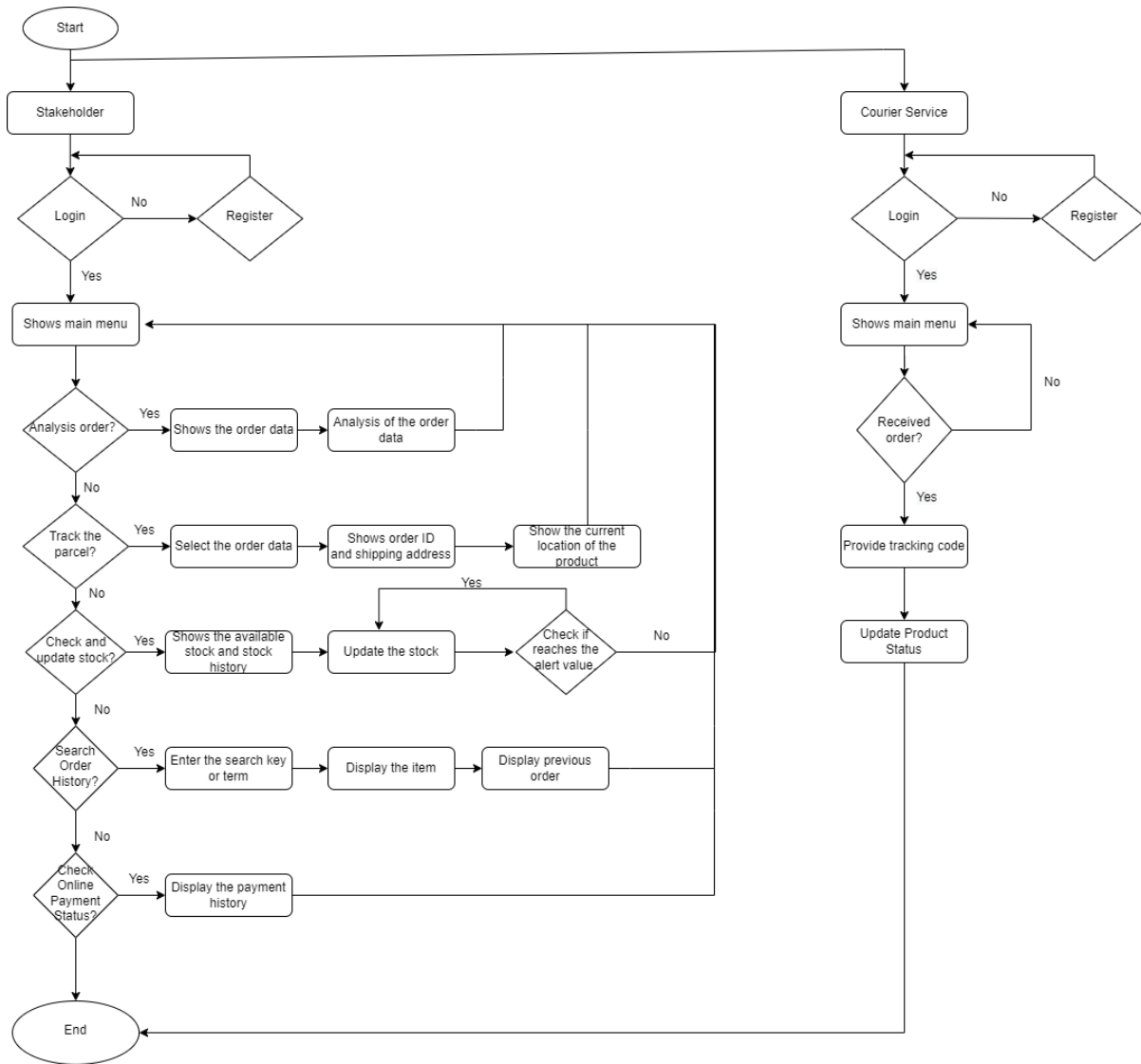
Here are the scenarios and workflow of current business process for stakeholder:

1. Login to the system.
2. Customizing according to preference.
3. Some options are displayed on the main menu.
4. Option for choosing to do the analysis of the order data.
 - 4.1 Views on the order data from list
 - 4.2 Analysis process start
5. Option for track the parcel
 - 5.1 Select the order data that want to track
 - 5.2 Displays the order ID based on the order data selected
 - 5.3 Displays the shipping address and product details
 - 5.4 Displays the current location where the product was located.
6. Option for checking and updating the stock
 - 6.1 Displays the available stock and previous stock history of the order data.
 - 6.2 Allow to update the stock even though the stock still enough
 - 6.3 Check whether the stock reaches the alert value, if reaches then it was compulsory for the stakeholder to update the stock to maintain the stock.
7. Option for searching the order history
 - 7.1 Stakeholder required to enter the search key by key in the term.
 - 7.2 System will display the related item to the stakeholder.
 - 7.3 System will also display the previous order made by customer.
8. Option for checking the online payment status
 - 8.1 Display the payment history done by customers.

Here are the scenarios and workflow of current business process for courier:

1. Login to the system.
2. Customizing according to preference.
3. Check whether the courier received an order.

4. If received, the courier should provide a tracking code.
5. While the parcel is in transit or on delivery, the courier will update the current status of the parcel including the condition and the location of the parcel.



AS-IS System Workflow

3.2 Functional Requirement (Input, process, output)

3.2.1 Context Diagram

PROCESS	INPUT	OUTPUT
Minisite.com System	Product Information Order Information Payment Information Sales Information Courier Information	Item Respond Order history as preferences Valid payment details Courier location

3.2.2 Level 0 Diagram

PROCESS	INPUT	OUTPUT
View Product List	Product Information Stock Information	Item Response Item Info Inventory updated
View Order List	Inventory updated Sorting preferences Order Status	Order history by requested Customer Info Valid Order Valid Status
View Payment Status	Order history by requested Valid Status Payment Information Valid Online Payment Status	Online payment details Sales Info
View Sales Report	Sales Info Valid order list Address order	Profit earned Profit earned by each product Total sales for a product
View Courier Status	Tracking Number	Live location about the parcel

3.2.3 Level 1 Diagram

3.2.3.1 Process 1: View Product List

PROCESS	INPUT	OUTPUT
Search Item	Item Information	Item Info Item remaining number of stocks
Response Item	Item remaining number of stocks Item Info	Item Response
Update Inventory	The number of purchases of new stocks	Order Information New inventory number

3.2.3.2 Process 2: View Order List

PROCESS	INPUT	OUTPUT
Search order	Order Information New inventory number	Order Info
Response Order	Order Info Order Number	Order Status Customer information Preference(Date/Month/Period)
Sort Order	Preference (Date/Month/Period)	Order Infor by preferences

3.2.3.3 Process 3: View Payment Status

PROCESS	INPUT	OUTPUT
Check Payment Status	Payment Status Valid Online Payment Status	Online Payment Details Sales Info
Receive Online Payment	Online Payment Order Infor by preferences	Customer Info Valid Status Payment Status

3.2.3.4 Process 4: View Sales Report

PROCESS	INPUT	OUTPUT
Check The Total Sales Report	Online Payment Details Sales Info Customer Info Valid Status	Sales Info Profit Earned
Check Profit Earned By A Product Report	Sales Info	Sales Info Profit Earned
Check Total Sales By A Product Report	Sales Info Profit Earned	Total sales Info Tracking Number Reference Number

3.2.3.5 Process 5: View Courier Status

PROCESS	INPUT	OUTPUT
Check the product delivery path	Total sales Info Address order Tracking Number Reference Number	Product Pathway Status

3.3 Non-Functional Requirement

Website for minisite: <https://minisite-utm.mystrikingly.com/>

3.3.1 Performance

- The performance grade for the minisite's website is 76 which is grade C.
- The page size of the minisite's website is 4.2MB.
- The load time of the minisite's website is 4.13s.
- The total requests of minisite's website are 181 requests.

3.3.2 Security

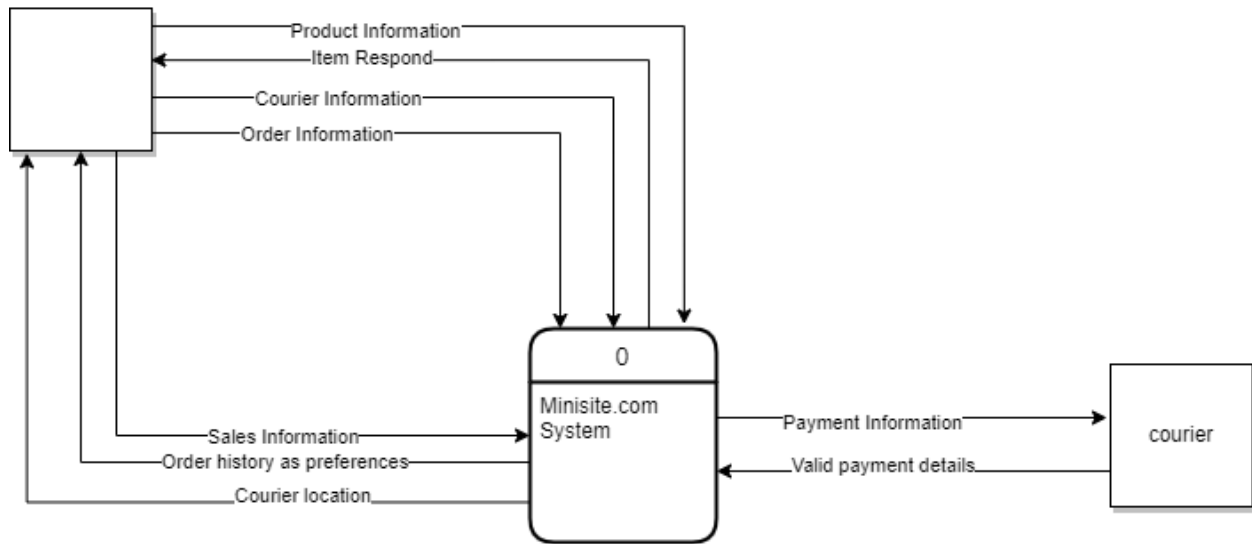
- Online payment transaction of any package should be considered as safe and secured.
- The personal information of the client need to be secured.

3.3.3 Compatibility

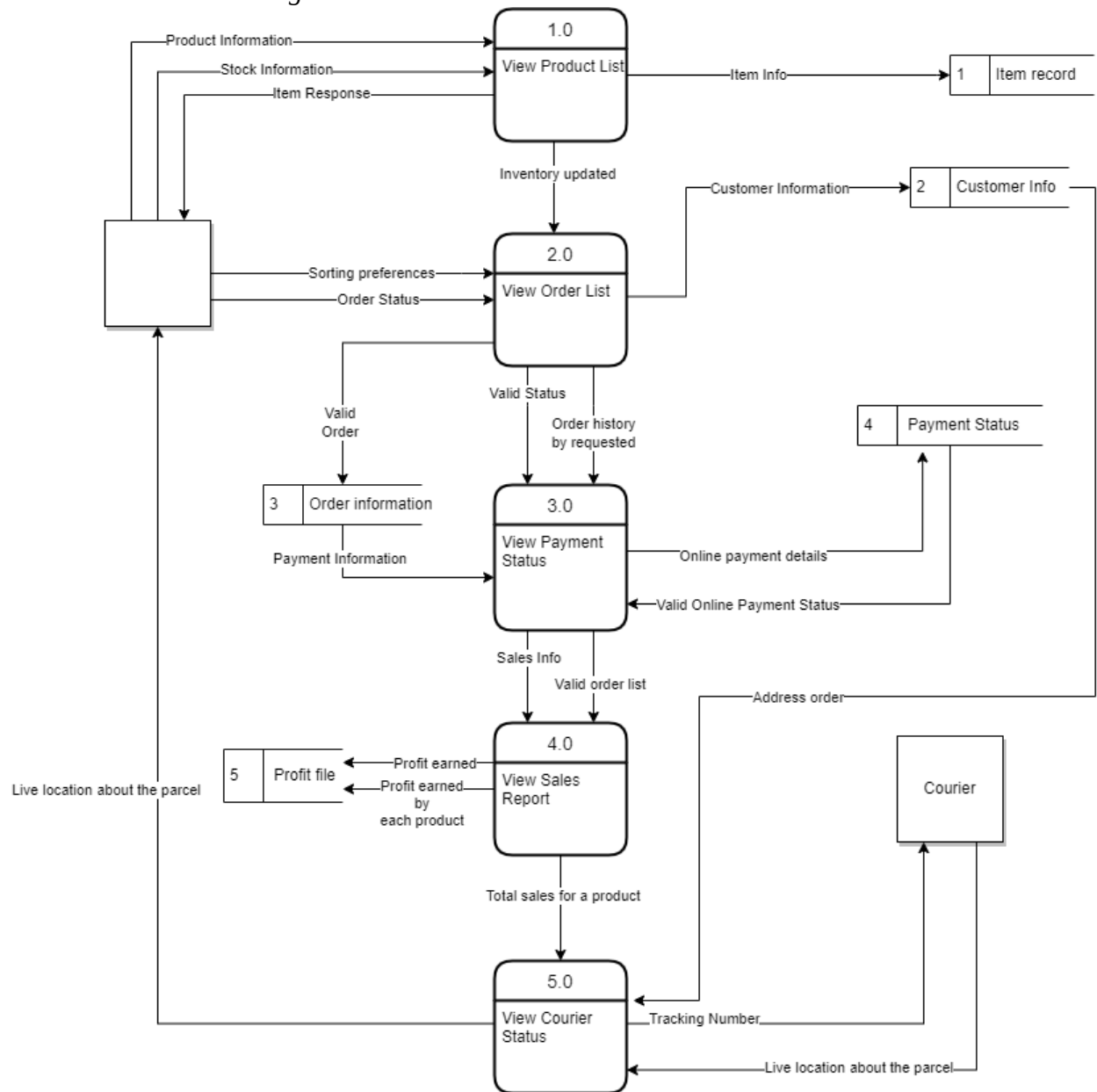
- Operating system of the device should be Windows 8/macOS High Sierra 10.13 or later
- Processor should be Intel Pentium 4 or later
- A minimum memory of 2GB is required.
- Internet connection is required.

3.4 Logical DFD for AS-IS System

3.4.1 Context Diagram

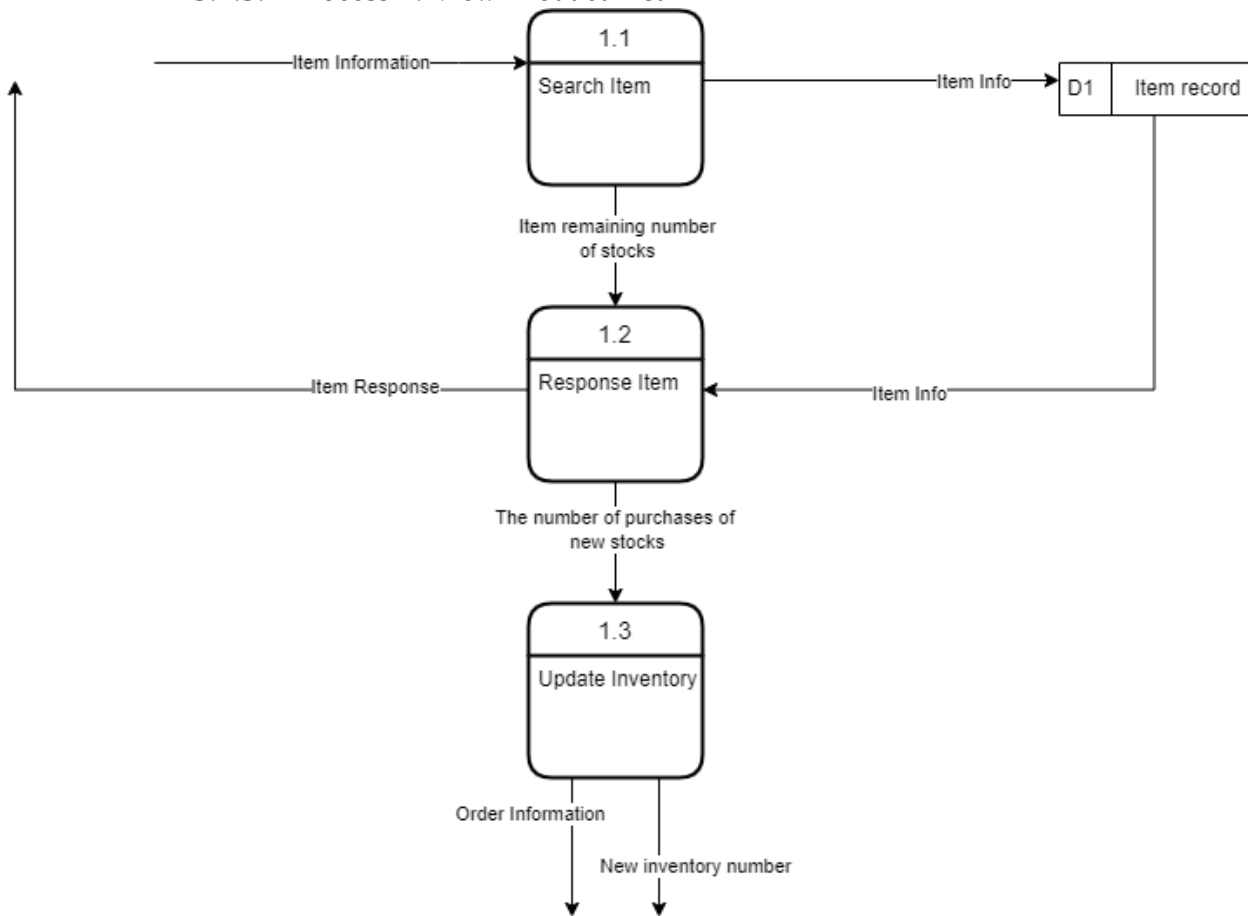


3.4.2 Level 0 Diagram

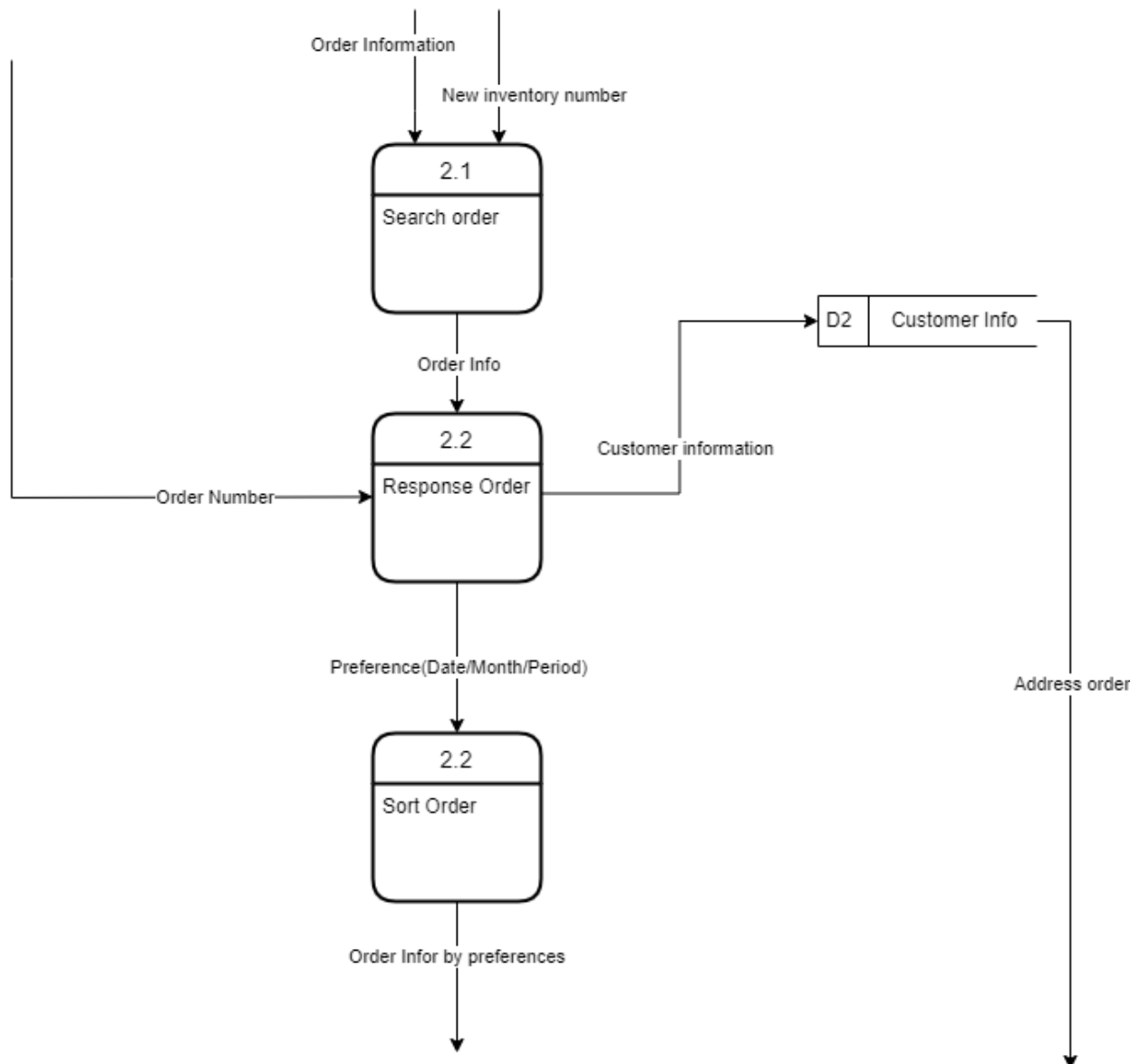


3.4.3 Level 1 Diagram

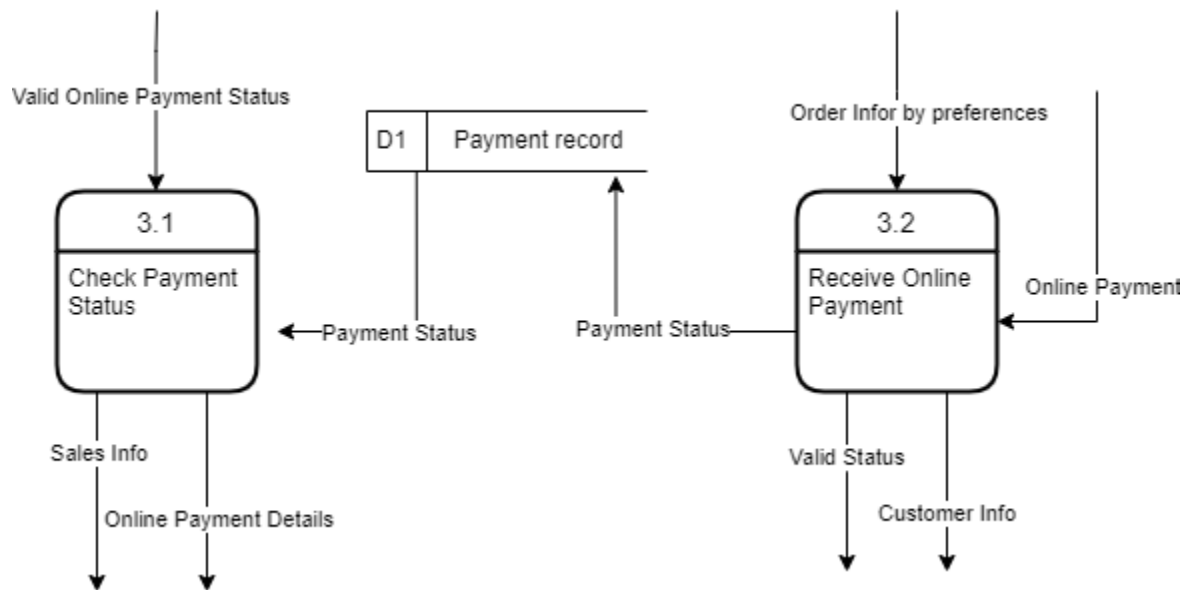
3.4.3.1 Process 1: View Product List



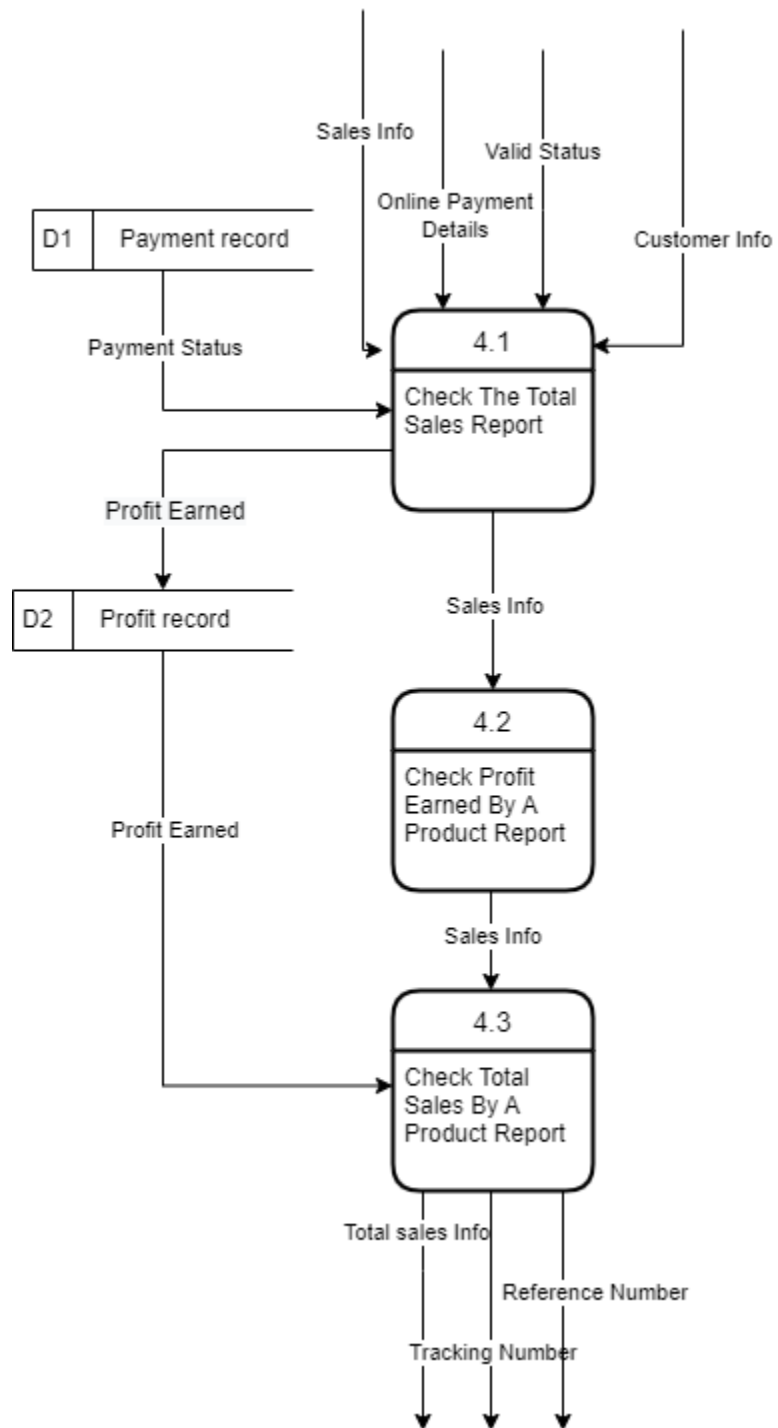
3.4.3.2 Process 2: View Order List



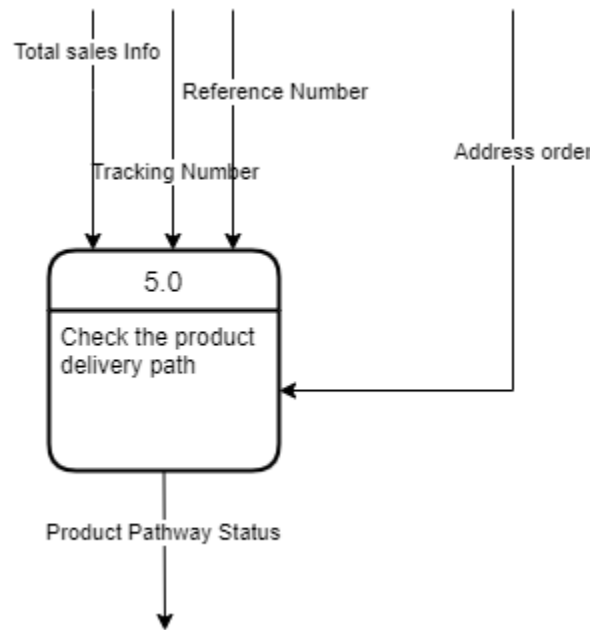
3.2.3.3 Process 3: View Payment Status



3.3.3.4 Process 4: View Sales Report



3.3.3.5 Process 5: View Courier Status



4.0 Summary of Requirement Analysis Process

The AS-IS ordering system of Richmine Trading Sdn. Bhd. has several parts that are done manually. These can be changed to automate so that it is more efficient and minimise the human mistake. Some facilities also can be added to the system to decrease the workload.

The function of the To-Be system are as follows:

1. Automated sales analysing
2. Database for order information and product information
3. Parcel's live tracking progress
4. Automated check stock
5. Sortation of sales record
6. Database for payment information